ROBLOX STUDIO EXPERT SKILLS · TEAM SKILL 3 OF 14

Luau Engineering

Gameplay programmer, Luau specialist

`scenario-roblox-luau`

Writes and reviews Luau that stays correct under Deferred events, type-checks on both solvers, frees what it creates, and keeps an optimization only when measured.



Live playtest capture in Studio 0.743 from the Luau test place, where 17 live tests pass.



Sawtooth memory that falls back is healthy; a leak never comes down. Andrew, Roblox Learn, 00:05:57.

USE IT WHEN

- Replacing wait() or spawn(), or handlers run in the wrong order
- 'attempt to yield across metamethod/C-call boundary' or --!strict type errors
- Server memory keeps growing, or 'make this script faster' with native or Actors

WHAT THE EXPERTS INSIST ON

- task.spawn runs the body up to its first yield before the caller continues; use task.defer for 'after this step'.
  Crusherfire (scripting educator, course author)
- Write Deferred-safe code: Deferred events alone cut one shipped game's frame time from 6 to 8 ms to 4 to 5 ms.
  Drew Hamilton (studio developer, RDC 2026)
- A leak is memory that never comes down; memory that grows and falls back is not a leak. Andrew (Roblox Learn)
- Scripts under one Actor run in series: use one Actor per independent unit, and write only after task.synchronize().
  Suphi Kaner (scripting educator)
- Add cleanup on the line that creates the connection or object, so nothing outlives its owner. B Ricey (Roblox educator)

WORKFLOW, WITH THE GATE BEFORE EACH NEXT STAGE

1. **Audit what exists.** **Gate:** 0 error findings; every warning fixed or justified in one line.
2. **Fix the idioms.** **Gate:** task.* only, no yields in frame handlers; audit clean, behavior unchanged.
3. **Type pass.** **Gate:** 0 errors on the old and the new type solver.
4. **Ownership and cleanup.** **Gate:** 0 objects alive after release; heap flat over 20 or more cycles.
5. **Probe order-dependent behavior.** **Gate:** Scheduler and signal probes recorded on Server and Client in one playtest.
6. **Optimize only with numbers.** **Gate:** Native 1.2x or more, parallel 1.5x or more, else revert.
7. **Hand off with evidence.** **Gate:** Every claim has a number or a test name, with build and date.

KEY NUMBERS

task.delay(0) resume	**0.07 ms from PreSimulation, 16.42 ms from Heartbeat**
task.wait(0.25) inside a Heartbeat handler	**16 threads stacked in 1 s**
Native speedup, typed Vector3 loop	**4.9x; untyped parameters 0.96x**
Parallel speedup, 1 vs 8 Actors	**7.6x (144.5 ms to 19.0 ms), 32 jobs**
SharedTable plain += 1 from 8 Actors	**79% of 160,000 updates lost; increment lost 0**
Memory budget	**about 6.25 GB server; about 2 GB phone, 700 MB for OS**

CHECKS THE AGENT RUNS ON ITS OWN WORK

- rbx_luau.typecheck: 0 errors on both solvers, with the project sourcemap.
- rbx_luau.audit 0 errors; toolkit Audit finds no legacy scheduler or deprecated API.
- Leak.retained alive 0 and a flat Leak.slope after every cleanup fix.
- Bench and Parallel ratios recorded before keeping native code or Actors.

HANDOFFS

Receives Module layout, cleanup library and signal modules from architecture; session state keyed by player from data.

Delivers BufferCodec schemas to networking; bench, parallel and leak results to performance; the type gate to tooling CI.

TOOLKIT RBX_LUAU.PY

- `typecheck()`: luau-lsp on old and new solvers, with sourcemap
- `audit()`: offline Lune scan for scheduler, yield, leak mistakes
- `run()`: Studio chunk with Probe, Leak and Bench kits

MAIN EXPERT SOURCES

Crusherfire
scripting educator, course author, The GREATEST API EXPLAINED: Roblox Task Library

Andrew (Roblox)
Roblox Learn, official channel, How to find and fix memory leaks on Roblox

Suphi Kaner
scripting educator, Multi Thread, Parallel Scripts, MicroProfiler: Roblox Scripting Tutorial

Drew Hamilton, Jason Knight, Brian Neeler
studio developer, Roblox engineer, Roblox PM, Mastering Server Authority, RDC 2026

STATUS

Tests offline suite passes, both type solvers clean

In app 17 live tests pass in Roblox Studio 0.743

G1 code architecture (shared with architecture)

without skills	10/15
with skills	15/15